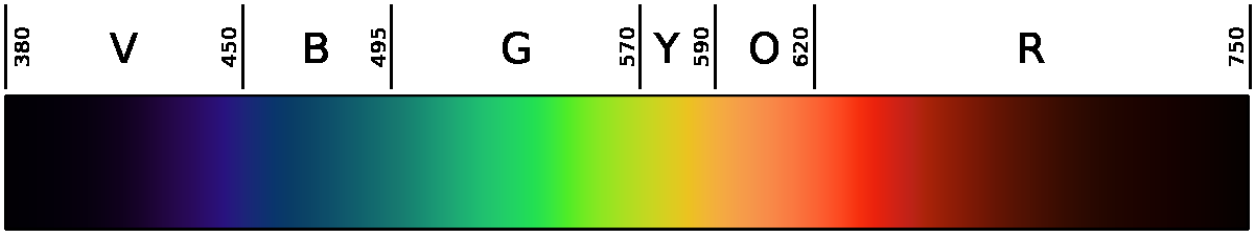
Ultraviolet



Ultraviolet is the part of the spectrum shown on the left side of the picture above as black—because humans cannot see light of such short wavelength (or high frequency). Many animals such as some insects, some reptiles, crocodiles, salamanders, and small birds can see things that reflect this light.

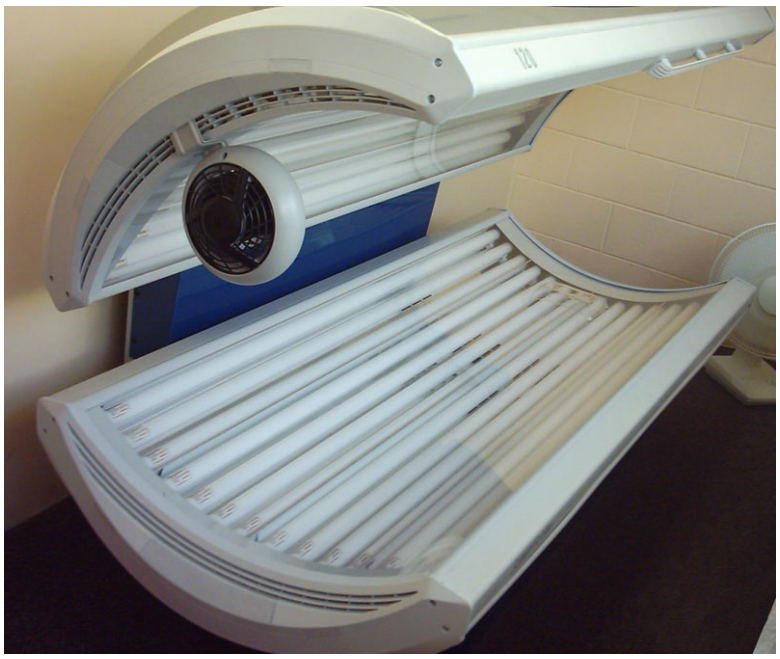
The name 'ultraviolet' uses the ancient Latin prefix *ultra-*, which means 'behind', 'beyond', or 'far'. Ultraviolet is beyond the visible violet light in terms of frequency, wavelength, and energy. Its wavelengths are between about 10 nanometers (nm) about 400 nanometers.

UV is a common abbreviation of ultraviolet, mainly used in technical contexts.

Frequency and wavelength are closely related. The equation that shows this relationship is: $\nu = c/\lambda$ (nu, i.e. frequency = speed of light ÷ lamda, i.e., wavelength), so saying that something has a short wavelength is equivalent to saying that it has a high frequency.

Ultraviolet radiation

Ultraviolet light can damage or kill cells. Any electromagnetic radiation (light) that has a wavelength shorter than 450 nm may cause trouble. Therefore humans that live in places with more ultraviolet light have adapted by getting darker skins. Pigments absorb the ultraviolet radiation, so it does not get through the skin to kill or injure cells inside. Injury to the skin by ultraviolet is called "sunburn."



Indoor tanning device

The only difference between violet light and ultraviolet light is the wavelength (or the frequency). The only difference between ultraviolet light and x-rays is also the wavelength or frequency. Ultraviolet is beyond violet, x-rays are beyond ultraviolet, and cosmic rays are beyond x-rays. Cosmic radiation can also be called gamma rays, or gamma radiation.

Technically, electromagnetic waves with a frequency of 789 terahertz or more are called ultraviolet. This corresponds with a wavelength from 380 nanometers down to approximately 5 nanometers. A single light quantum or photon of ultraviolet carries an energy around some 3.26 electron volt. The higher a photon's frequency, and so the shorter its wavelength, the more energy a photon carries.

Although the air of the Earth is transparent to a broad range of ultraviolet, some ultraviolet sunlight is absorbed at a very high altitude by the ozone layer. Recent and ongoing destruction of ozone in high altitudes caused by human influence—mostly by industrial chemicals and air travel—has greatly increased the amount of ultraviolet light reaching the Earth's surface. This, in turn, has increased the risk of skin cancer to mankind, and this risk will only increase with time unless the ozone layer is better protected.

Ultraviolet wavelengths less than 200 nanometers, X-rays, and gamma rays, are collectively called ionizing radiation since the energy in any such light quantum is high enough to 'kick' an electron out of an atom. This is why these kinds of radiation are dangerous to life.

Ordinary glass does not let radiation through if its wavelength is less than 200 nanometers, so it acts as a shield against the more dangerous range of ultraviolet light, but some special sorts of glass do not shield as well, including many car windows.

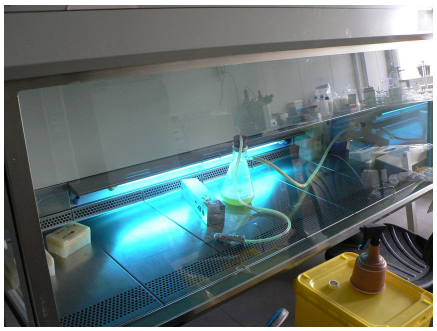
One use for ultraviolet radiation is tanning. Use of tanning devices can cause skin cancer because ultraviolet goes through the skin and causes destruction to cells causing sunburn.

Because of the destructive power of ultraviolet light, it can be used to kill germs. Sunlight is a powerful disinfectant.

Ultraviolet lamp

An ultraviolet lamp is one that emits light mostly at a frequency higher than violet, called ultraviolet. These lamps are often used to kill microbes (germs). They can be very powerful, so the people who work around them when they are turned on may need to wear protective glasses and keep their skin covered to avoid injury.

In the laboratory pictured, ultraviolet lights are turned on when the workers are gone so that anything on the table surface will be killed. Besides ultraviolet light, which makes up most of the light produced by these lamps, there is also a little violet and blue light. So people can easily tell when the ultraviolet lamps are turned on.



An example of an ultraviolet lamp

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